

POLTAVKA, Russian Federation

WMO: 287860

Lat: **54.3561N**Lon: **71.7661E**Elev: **125**StdP: **99.83**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.9	-36.5	0.1	-33.5	-33.8	0.2	-30.8	9.2	-14.0	8.4	-11.9	1.5	180	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	31.2	18.5	29.2	18.1	27.4	17.6	20.6	27.2	19.7	26.1	18.9	25.2	3.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	23.1	17.5	12.7	22.3	16.5	11.9	21.6	59.8	27.3	56.9	26.0	54.3	25.3	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.7	6.9	-36.0	34.1	3.7	1.7	-38.6	35.3	-40.8	36.3	-42.8	37.3	-45.5	38.6		
(5)			-36.3	22.1	3.6	1.0	-38.9	22.8	-41.0	23.4	-43.1	24.0	-45.7	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-17.6	-15.5	-7.4	4.5	12.7	18.3	19.4	17.5	11.4	3.9	-6.6	-13.9	(6)
(7)		DBStd	14.60	7.66	7.28	6.33	6.25	5.18	4.50	3.68	4.11	4.59	5.12	7.52	7.81	(7)
(8)		HDD10.0	3801	856	714	541	184	32	2	0	1	36	198	497	741	(8)
(9)		HDD18.3	6021	1114	948	799	415	184	56	32	67	212	448	747	999	(9)
(10)		CDD10.0	998	0	0	0	19	116	251	293	233	79	8	0	0	(10)
(11)		CDD18.3	177	0	0	0	0	10	55	67	40	5	0	0	0	(11)
(12)		CDH23.3	1838	0	0	0	6	169	581	606	404	72	1	0	0	(12)
(13)		CDH26.7	560	0	0	0	1	39	199	178	126	18	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.2	3.3	3.7	3.7	3.6	2.9	2.6	2.9	3.3	3.5	3.5	(14)	
(15)	Precipitation	PrecAvg	365	17	15	16	23	30	47	64	48	24	30	29	22	(15)
(16)		PrecMax	569	38	31	40	60	88	116	116	95	62	79	75	43	(16)
(17)		PrecMin	223	2	2	1	4	4	19	13	15	4	2	7	5	(17)
(18)		PrecStd	79	9	7	11	14	17	22	28	23	14	19	15	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	0.8	7.1	23.7	30.3	33.3	33.9	32.9	29.0	20.4	7.5	1.5	(19)
(20)			MCWB	-0.1	-0.1	4.1	12.7	15.2	18.5	19.9	18.7	15.4	11.4	5.6	0.3	(20)
(21)		2%	DB	-1.8	-0.7	3.4	20.0	27.3	31.1	30.4	29.8	25.3	17.3	5.7	0.1	(21)
(22)			MCWB	-2.6	-1.6	1.7	10.7	14.5	18.3	19.1	18.6	14.1	10.3	4.1	-0.5	(22)
(23)		5%	DB	-3.8	-2.5	2.1	17.3	25.1	29.0	28.4	27.5	22.1	14.4	4.0	-1.3	(23)
(24)			MCWB	-4.4	-3.2	0.8	9.9	14.0	17.9	19.0	17.9	13.4	8.7	2.5	-2.0	(24)
(25)		10%	DB	-6.0	-4.7	1.0	14.3	22.7	26.6	26.7	25.4	19.2	11.6	2.1	-3.6	(25)
(26)			MCWB	-6.6	-5.4	0.0	8.5	13.0	17.2	18.7	17.3	12.2	7.7	0.9	-4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.1	4.1	13.7	17.1	21.4	21.9	21.0	16.7	12.8	6.2	0.7	(27)
(28)			MCDWB	0.5	0.7	6.9	21.1	24.6	28.8	28.1	28.2	25.5	18.9	7.2	1.1	(28)
(29)		2%	WB	-2.4	-1.5	1.9	11.8	15.7	20.0	20.9	19.8	15.3	11.1	4.2	-0.5	(29)
(30)			MCDWB	-1.8	-0.7	3.1	18.1	24.8	27.4	27.3	26.4	22.6	15.5	5.5	0.1	(30)
(31)		5%	WB	-4.4	-3.2	1.0	10.2	14.7	19.0	20.0	18.9	14.1	9.3	2.4	-2.0	(31)
(32)			MCDWB	-3.7	-2.6	1.9	16.6	23.3	26.3	26.2	25.3	20.8	13.6	3.8	-1.2	(32)
(33)		10%	WB	-6.6	-5.4	0.0	8.7	13.7	17.9	19.3	18.0	12.9	7.8	1.0	-4.3	(33)
(34)			MCDWB	-6.1	-4.7	1.0	14.0	21.2	25.0	25.2	24.1	18.3	11.3	2.0	-3.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.0	8.2	9.7	12.2	11.2	10.2	10.7	10.3	8.0	6.4	7.3	(35)
(36)			MCDBR	8.2	7.5	7.0	14.7	15.9	14.5	13.0	14.6	14.9	13.3	6.5	7.5	(36)
(37)			MCWBR	8.1	7.2	5.9	7.6	6.7	5.5	5.0	5.8	6.7	7.6	5.5	7.3	(37)
(38)		5% WB	MCDBR	8.1	7.4	6.5	13.5	14.0	12.4	10.9	12.3	13.1	11.6	6.1	7.5	(38)
(39)			MCWBR	8.0	7.1	5.8	7.6	7.0	5.3	4.9	5.7	6.4	7.2	5.5	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.317	0.375	0.409	0.406	0.410	0.435	0.430	0.372	0.352	0.323	0.289	(40)	
(41)		taud	2.049	2.018	1.951	2.131	2.220	2.295	2.212	2.190	2.314	2.253	2.117	2.021	(41)	
(42)		Ebn at Noon	622	737	772	811	839	839	806	783	788	703	572	520	(42)	
(43)		Edh at Noon	71	104	140	135	131	123	131	126	98	83	67	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.05	2.16	3.69	4.66	5.71	6.07	5.58	4.61	3.29	1.70	0.83	0.68	(44)	
(45)		RadStd	0.07	0.08	0.33	0.43	0.46	0.35	0.41	0.36	0.24	0.23	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page